E

(a) How the surrogate check works

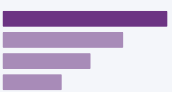
1. Start from each patient's out-of-bag risk score.



2. A random forest learns that score from clinical (21 BLCA, 25 BRCA) and first-order pathomic features (504, 514).



3. Rank features and compartments by importance (impurity).



4. Two checks: does the order survive a second estimator, and does each feature predict observed PFI after false-discovery-rate control?

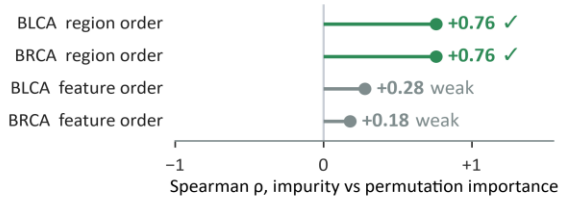
- permutation importance
- Cox vs PFI, FDR

WHY IT MATTERS

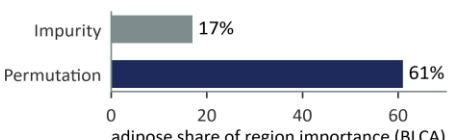
The surrogate explains the model's score. Only features that pass the outcome test say something about the disease.

(b) What the audit found

A second estimator: region order holds



Same leading region, but its share is not robust



Against outcome, no feature survives correction

